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# THE COST OF VOLUNTARY EXCHANGE: ALLOCATION AND SETTLEMENT UNDER SEPARATE PUBLIC ACCOUNTS

An informed authority chooses allocations and voluntary settlement schedules under separate public accounts. We characterize implementation losses as functions of standing capacity, temporary reserves, and approval costs. Conserving technologies recycle capacity; executable productive releases expand it. With empty accounts, optimized losses have square-root order under conservation and linear order with a logarithmic factor under productivity. Positive standing capacity changes these limiting rates while preserving a quantitative gap in approval requirements. This gap determines when productive releases reverse the choice between joint and sequential settlement. The model retains complete information, divisible irreversible adjustments, balanced compensation, and unvalued residual public balances.

KEYWORDS: Voluntary exchange, public accounts, bilateral clearing, reserves, settlement costs.

JEL CLASSIFICATIONS: D47, D61, H41, H55, J26.

## 1. INTRODUCTION

An informed public authority chooses an allocation and a voluntary settlement schedule under separate public accounts. A transaction can consume capacity before another transaction releases it. This execution constraint changes the authority's choice between sequential approvals and joint settlement, even when each institution chooses its own allocation. Conserving transactions reuse a bounded stock; executable productive releases expand the capacity available for later approvals. We derive sharp implementation-loss bounds and identify when this difference reverses the preferred institution. Balanced compensation secures consent within every approval, while account feasibility governs its execution. The resulting comparison determines how standing capacity substitutes for temporary reserves and joint commitment.

The analysis begins with the fully informed planner and its implementation through unrestricted voluntary trade. We then impose separate-account feasibility after every bilateral approval. With empty accounts, conserving technologies incur optimized losses of order $\sqrt{\rho\tau}$, whereas executable productive releases yield order $\tau \log(\rho/\tau)$, where $\rho$ prices

temporary reserves and $\tau$ prices approvals. Positive standing balances change both limiting rates to linear order in $\tau$ when those balances are fixed. The economically relevant difference then lies in the dependence on standing capacity: inverse under conservation and logarithmic under productivity. A uniform comparison identifies a range in which existing balances sustain productive implementation at the minimum-loss order without rented reserves, while conserving implementation benefits from additional reserves. The associated welfare gap determines when productive releases change the preferred settlement institution. Throughout, the informed authority selects allocations and schedules; adjustments are divisible and irreversible, and compensation is enforceable and balanced. The model contains no decentralized bargaining.

*Contribution hierarchy.* The principal result links the resource technology to optimized implementation losses, jointly accounting for standing capacity, rented reserves, and approval costs. Theorem 9 establishes the comparison with positive initial balances; Theorem 6 gives its empty-account benchmark. The exact reserve–approval frontier identifies the mechanism, and the general execution conditions extend it beyond the numerical family. These bounds have two institutional implications. They identify when existing capacity is sufficient to implement the planner allocation at the optimal loss order without rented reserves, and when the cost of joint commitment changes the preferred settlement rule. A common-fee interval establishes the latter comparison after both institutions choose their allocations. Corollary 14 also solves the institutional decision at the original fixed prices: a strictly positive standing-capacity interval supports the reversal, and a higher capacity cutoff makes sequential settlement preferable under both technologies. Table I organizes the results around this argument.

*The approval frontier and its boundary.* A blocked start and a strictly positive non-expanding weighted stock imply inverse-reserve approval requirements. An executable productive sequence supports geometric expansion and logarithmic requirements when its releases also permit completion at the planner allocation. We state both sets of conditions directly and provide a finite family of linear programs that constructs the productive sequence from the resource matrix, preferences, caps, and planner value. Section 4.5 extends the comparison uniformly to positive standing balances. The resulting bounds separate three policy margins: using standing capacity, authorizing temporary reserves, and paying for

TABLE I

CONTRIBUTION HIERARCHY

| Role | Economic content | Results |
|---|---|---|
| Central result | Standing capacity and productive releases determine optimized implementation losses and the value of temporary reserves. | Theorems 9 and 6 |
| Exact mechanism | Conservation and productive releases generate inverse and logarithmic approval requirements. | Propositions 2 and 13; Theorem 4 |
| Institutional implication | Commitment thresholds identify an endogenous-allocation reversal at common prices. | Corollaries 10 and 14; Theorem 12 |
| Economic margins | Allocation-dependent fees, distorted quantities, binding participation, and discrete decisions. | Propositions A.5and A.6; Corollary 15 |
| Foundation | Balanced implementation and exact recovery with a repayable reserve. | Section 2.3; Theorem A.2 |
| Application | Priority changes the number and identity of water trade requests served by a fixed capacity. | Section 6; Appendix M |

joint commitment. The exact constants and parity terms in the three-account family sharpen this general comparison.

*Mandate and extensions.* The core sets $B \equiv 0$: employee surplus is maximized subject to separate public protections, with residual balances carrying no additional welfare value. Implementation costs are financed from separately authorized operating resources and deducted from aggregate surplus. A curved-benefit example shows how sequential costs reduce the selected quantity; a participation example shows how employee consent raises the reserve required by a feasible physical schedule. The institutional reversal also holds over a stated interval of allocation-dependent fees and for a fixed adjustment grid with positive initial balances. Appendix Kstudies valued public balances and the associated financing conditions. Appendix Lderives discrete reserve floors and convergence to the divisible frontier.

*Administrative application.* Section 6 examines a published water trade queue with positive initial capacity. The same 203 submitted requests produce 98 successful applications under recorded random priority, 106 under arrival order, and a maximum of 159 under smallest volume first. Executed water volume changes by less than $0.1\%$. The application

isolates the allocation consequences of priority conditional on the submitted requests and other account requirements. The exact count benchmark and administrative replay give an observable institutional example of why high aggregate capacity use can coexist with substantial differences in access across applications.

*Relation to existing work.* The bilateral-exchange literature characterizes the relationship between pairwise improvements and efficiency (Feldman, 1973, Madden, 1978, Goldman and Starr, 1982, Fisher, 1989). We study the execution cost of voluntary improvements when separate public accounts constrain every intermediate allocation. Continuous-net theory provides finite-reachability characterizations and divided-execution constructions (Júlvez et al., 2003, Fraca and Haddad, 2015). Our economic comparison optimizes the allocation and its voluntary implementation together. Preferences and balanced compensation determine admissible atomic approvals; account protections determine their executable order. The reserve frontier prices this interaction and yields thresholds for institutional choice, including with positive standing capacity. Decomposing an atomic approval into separate activity moves can change consent, resource requirements, and implementation cost.

Settlement economics establishes that the organization of execution affects resource use and welfare. Kahn and Roberds (2001) derive the cost of gross settlement from credit constraints and analyze the role of intraday credit and collateral. Kahn et al. (2003) connect the underlying trading economy to gridlock, collateral requirements, and settlement delay. Martin and McAndrews (2008) study how queue design changes incentives and welfare, while Atalay et al. (2010) compare settlement outcomes with the planner's allocation. Related work studies settlement institutions in trading economies (Koeppl et al., 2008). Our mechanism concerns how voluntary execution replenishes the particular public capacities needed for subsequent approvals. Its economic consequence is a change in optimized implementation losses and in the authority's willingness to pay for joint authorization. The conserving cost balance follows the inventory logic of Baumol (1952); productive releases change the rate at which additional capacity reduces the approval burden. The authority chooses the allocation and schedule under complete information. Private-information and computational-complexity questions are developed separately.

## 2. THE PLANNER AND UNRESTRICTED TRADE

### 2.1. *Preferences, accounts, and the settlement mandate*

There are finite sets of demanders $D$ and suppliers $S$. Demander $i$ chooses $a_i \in [0, \bar{a}_i]$ and receives benefit $U_i(a_i)$; supplier $j$ chooses $b_j \in [0, \bar{b}_j]$ and incurs cost $C_j(b_j)$. Roles are fixed and publicly verified. Let $z = (a, b)$ and let $X$ be the product of these intervals, with positive finite caps. The baseline is $z = 0$. Benefits and costs are continuously differentiable on neighborhoods of their intervals, benefits are concave, and costs are convex and nondecreasing. Normalize $U_i(0) = C_j(0) = 0$. The frontier and cost results specialize to positive linear values and costs, $U_i(a_i) = v_i a_i$ and $C_j(b_j) = c_j b_j$.

There are $K$ public resource accounts. Their initial capacities are $R \geq 0$, and $M$ is the verified linear resource-effect matrix. A positive entry of $Mz$ releases capacity; a negative entry consumes it. The feasible set is

$$\mathcal{F}^S(R) = \{z \in X : R + Mz \geq 0\}. \tag{1}$$

Inequalities between vectors are coordinatewise. Rows can distinguish dates, grades, or organizational units whose authorizations cannot be combined. For one demander, the retirement interpretation has pension, payroll, and position accounts:

$$R + Mz = \begin{pmatrix} R^P - \kappa_d a + \sum_j \kappa_j b_j \\ R^W + \omega_d a - \sum_j \omega_j b_j \\ R^H + a - \sum_j b_j \end{pmatrix}, \qquad \kappa_d, \omega_d, \kappa_j, \omega_j > 0. \tag{2}$$

The pension row is a specified ceiling on incremental pension exposure, not an agency-held retirement fund. Payroll effects must be recognized within the relevant authorization, and position units represent compatible service time. The institutional technology grants authority to certify and reuse these releases.

Compensation $t_h$ is money received by employee $h$. Employee utilities are $U_i(a_i) + t_i$ and $-C_j(b_j) + t_j$. Compensation is exactly balanced when $\sum_h t_h = 0$; these transfers do not enter $M$. They do not amend an appropriation or create a position. Define employee surplus by

$$W(z) = \sum_{i \in D} U_i(a_i) - \sum_{j \in S} C_j(b_j) = \sum_h f_h(z_h). \tag{3}$$

**Settlement mandate.** The core sets $B \equiv 0$: the authority maximizes employee surplus subject to every separate public-account constraint, with no additional welfare weight on residual balances and no net public payment to employees. Information is complete, adjustments are divisible and irreversible, and compensation is enforceable and balanced. Every account remains nonnegative after each approval. Temporary reserves must be returned; original balances may be consumed. Implementation expenditures come from separately authorized operating resources.

The fully informed benchmark under this mandate is

$$V^S(R) = \max_{z \in \mathcal{F}^S(R)} W(z). \tag{SP}$$

Compactness and continuity give an optimum; the baseline makes its value nonnegative. This mandate fixes public protections as constraints. A broader appraisal that values residual capacity is developed in Appendix K.

### 2.2. *The endogenous planner allocation in the productive family*

The central family has one demander, two suppliers, unit caps, zero initial balances, $v = 1$, $c_1 = c_2 = 1/10$, and

$$M_h = \begin{pmatrix} -2 & 3/2 + h & 5/2 \\ 1 & -1/2 & -3/2 \\ 1 & -1 & -1 \end{pmatrix}, \qquad 0 \le h < 1/2.$$

Only supplier 1's pension release changes with $h$. Preferences, caps, the other resource effects, and bilateral permissions stay fixed.

PROPOSITION 1—The endogenous planner optimum: *For each demander quantity $a \in [0, 1]$, the minimum supplier cost under the separate-account constraints is*

$$C_h^S(a) = \frac{1+h}{5(2+3h)}a, \qquad (b_1, b_2) = \left(\frac{a}{2+3h}, \frac{(1+2h)a}{2+3h}\right).$$

*The planner optimum and value are therefore*

$$z_h^* = \left(1, \frac{1}{2+3h}, \frac{1+2h}{2+3h}\right), \qquad V_h^* = 1 - \frac{1+h}{5(2+3h)}.$$

*The optimum is unique, and its residual balances are* $M_h z_h^* = (0, 0, h/(2+3h))^{\mathsf{T}}$.

PROOF: At a fixed $a$, pension and payroll feasibility imply

$$(3/2 + h)b_1 + (5/2)b_2 \geq 2a, \qquad -(1/2)b_1 - (3/2)b_2 \geq -a.$$

Multiplying these inequalities by $2/(2+3h)$ and $2(1-h)/(2+3h)$, respectively, and adding gives

$$b_1 + b_2 \geq \frac{2(1+h)}{2+3h}a.$$

The stated bundle attains this bound and leaves position capacity $ha/(2+3h) \geq 0$. Its quantities respect the unit caps. Both multipliers are positive, so equality requires both fiscal constraints to bind; their independent equations give the unique cost-minimizing bundle. Employee surplus at this bundle is $a[1-(1+h)/(5(2+3h))]$, whose coefficient is positive. Hence the unique optimum has $a = 1$. *Q.E.D.*

At $h = 1/4$, the optimum is $(1, 4/11, 6/11)$ and its value is $10/11$. The proportional target $(1, 1/2, 1/2)$ has value $9/10$ and leaves pension capacity $1/8$. It is feasible but not planner-optimal: it uses more supplier activity to obtain the same demander benefit. In general,

$$V_h^* - \frac{9}{10} = \frac{h}{10(2+3h)},$$

which is strictly positive for $h > 0$ and equals $1/110$ at $h = 1/4$.

### 2.3. *Unrestricted trade implements the benchmark*

An unrestricted agreement makes an entire feasible allocation and its compensation effective together. Each employee may retain the baseline, and there are no wealth constraints or transfer caps. At any allocation with $W(z) \geq 0$, choose $\alpha_h \geq 0$ summing to one and set $t_h = \alpha_h W(z) - f_h(z_h)$. Transfers balance and give each employee the nonnegative payoff $\alpha_h W(z)$. Thus unrestricted voluntary agreement implements a planner optimum.

The allocation also has competitive support. At account prices $p \geq 0$, employee $h$ chooses $z_h$ to maximize $f_h(z_h) + p^{\mathsf{T}} m_h z_h$, where $m_h$ is its column of $M$. Concavity and the polyhedral normal-cone condition at an optimum supply supporting prices with $p^{\mathsf{T}}(R + Mz^*) = 0$.

Fixed nonnegative rebates totaling $p^{\mathsf{T}}R$ make employee transfers exactly balanced. Each employee weakly prefers its choice to zero. Conversely, summing the maximizing inequalities establishes planner optimality. This standard welfare argument (Arrow and Debreu, 1954, Boyd and Vandenberghe, 2004) is proved, including valued public balances, in Proposition A.8of Appendix K.

For $M_h$, prices

$$p_h = \left(\frac{1}{5(2+3h)}, \frac{1-h}{5(2+3h)}, 0\right)$$

pay each supplier $1/10$ per unit and charge the demander $(1+h)/[5(2+3h)]$ per unit. The demander chooses one; the cost-minimizing supplier quantities clear the accounts and satisfy supplier optimality. The corresponding transfers leave the demander $V_h^*$ and both suppliers zero. These prices support the planner allocation through jointly effective agreements under complete information.

## 3. CONSTRAINED SETTLEMENT AND THE EXACT RESERVE FRONTIER

A joint package takes effect when its aggregate allocation satisfies the ledger. A sequential approval changes at most one demander and one supplier activity; demander-only approvals are permitted. Activities only increase. Each approval must leave every account nonnegative and give each participating employee a nonnegative incremental gain under exactly balanced compensation. With linear preferences, increments $(x, y)$ for pair $(i, j)$ satisfy this participation condition precisely when $v_i x \geq c_j y$. Previously settled releases can be reused. An independently cleared pair cannot use another pair's release; the sequential institution studied here expressly permits such reuse.

Let $V_T^{\text{seq}}(R)$ denote the largest employee surplus attainable from $R$ in at most $T$ such approvals. This notation alone imposes no repayment condition. In every institutional comparison below, a reserve $\epsilon r$ added to an original ledger $R^0$ must be returned, so the terminal allocation also satisfies $R^0 + Mz \geq 0$. Reserve authorization and return are not bilateral approvals. All costs below price this same convention.

Two implementation facts motivate the frontier. First, no nonzero bilateral approval is feasible from zero in $M_h$: supplier 1 requires $x \geq y$ from positions but $x \leq (3/4 + h/2)y < y$ from pensions; supplier 2 requires $x \geq 3y/2$ from payroll but $x \leq 5y/4$ from pensions. A demander-only move also consumes pension capacity. Second, a reserve positive in

every account permits exact recovery of an original positive-surplus planner optimum under nondecreasing concave benefits, convex costs with positive derivatives, and complete bilateral access. Theorem A.2proves this claim by decomposing the optimum into employee-voluntary bundles and executing sufficiently small rounds. The reserve covers within-round shortfalls and is returned at completion. Its full statement and proof are in Appendix A.

For $\theta \in (0,1]$, define the proportional target $z_\theta = (\theta, \theta/2, \theta/2)$ and let $N_h(\theta, \epsilon)$ be its minimum number of voluntary approvals from $\epsilon(1,1,1)$. Its surplus is $9\theta/10$ and its net release is $(h\theta/2, 0, 0)$, so the reserve is repayable. This target provides an exactly solvable intermediate stage. Proposition 1 shows why it is not the planner optimum when $h > 0$.

PROPOSITION 2—The exact reserve frontier with productive releases: *In the economy $M_h$, fix $\theta \in (0,1]$ and an integer $T \geq 2$. The least reserve per account that permits exact voluntary implementation of $z_\theta$ in at most $T$ bilateral approvals is*

$$\epsilon_T(h,\theta) = \begin{cases} \dfrac{h\theta}{6(G_T(h)-1)}, & 0 < h < 1/2, \\ \dfrac{\theta}{12(T-1)}, & h = 0, \end{cases} \tag{4}$$

*Here $\gamma_h = (1+2h)/(1-2h)$ and, for $n \geq 1$,*

$$G_{2n}(h) = \frac{1+\gamma_h}{2}\gamma_h^{n-1}, \qquad G_{2n+1}(h) = \gamma_h^n.$$

*An attaining schedule alternates suppliers, beginning with supplier 1. Every intermediate account balance is nonnegative, each transaction is voluntary under balanced compensation, and the reserve can be returned after the final approval. The formula for $h = 0$ is also the continuous limit as $h \downarrow 0$ for each fixed $T$ and $\theta$.*

PROOF SKETCH: A change of variables projects feasible balances into a two-dimensional conversion problem. In that relaxation, deleting moves without supplier activity and combining adjacent moves using the same supplier leaves an alternating word. Successive maximal conversions move between its boundary faces; the associated linear recurrence has multiplier $\gamma_h$ and the displayed parity term $G_T$. Bounding all alternating words by this recurrence gives necessity for physical schedules. Lifting the boundary construction back to activity increments attains the bound, has positive surplus at every approval, respects the activity

caps, and returns the reserve. At $h = 0$ the recurrence becomes additive. Appendix Bproves both bounds and verifies the attaining transfers and terminal balances. *Q.E.D.*

The frontier has inverse-reserve order at $h = 0$ and logarithmic order at every fixed $h > 0$. Conservation keeps the usable weighted stock proportional to the initial reserve. Productive rounds create capacity that later rounds reuse. Section 4.1 gives conditions for this distinction beyond $M_h$ and for completion at an endogenous planner optimum.

## 4. PRODUCTIVE RELEASES AND OPTIMIZED IMPLEMENTATION LOSSES

The authority authorizes reserve $\epsilon r$ at cost $\rho\epsilon$, with $\rho > 0$, and pays $\tau > 0$ for each approval. A joint package uses one approval and pays a fixed contingent-commitment cost $F$. Unless otherwise specified, $F$ is the same for every nonzero joint package, and both institutions have common verification cost $\mathcal{K} = 0$. The zero allocation costs nothing. Costs are paid from preauthorized operating resources and deducted from aggregate surplus; employee compensation remains balanced within each approval. If these costs consume a constrained account, that consumption must instead be included in the ledger. No exogenous reserve cap is imposed here.

*Authorization capacity and the institutional comparison.* An account balance authorizes specified adjustments within one planning episode. A reserve temporarily enlarges that authorization and is returned at the terminal allocation. Its price $\rho\epsilon$ measures the opportunity or administrative cost of making the reserve vector available. The retirement application interprets these balances as permissions to recognize and reuse pension, payroll, and position releases. The reserve composition $r$ is fixed.

We model two authorization technologies. Under routine approval, a bilateral order becomes effective when issued, compensation is enforceable, and each account officer checks the resulting ledger against the capacity then available. The authority can plan the entire sequence in advance. A planned future release becomes usable when the order producing it takes effect. Under joint authorization, the officers certify a common program whose debits and releases become effective together. The program requires a contingent authorization from each protected account and a common effectiveness instruction. Let $f_k \geq 0$ be the resource cost of preparing that contingent authorization for account $k$, and let

$f_0 \geq 0$ be the cost of activating the common instruction. With fixed administrative coverage, the additional program cost is

$$F = f_0 + \sum_{k \in \mathcal{A}} f_k,$$

where $\mathcal{A}$ is the fixed set of protected accounts. The ordinary approval cost $\tau$ covers the final authorization under either technology. The costs $f_k$ purchase conditional effectiveness across orders; the compensation contracts within each approved order are enforceable under both technologies. This administrative technology therefore separates the ability to plan a program from the authority to make its component orders jointly effective. Complete information removes uncertainty about gains and resource effects; it leaves these authorization requirements in place. In financial settlement, linking the completion of obligations is likewise an explicit design requirement (CPSS and IOSCO, 2012, Principle 12).

The cost $F$ is an avoidable resource cost of the program under consideration. When the required joint facility is already available at zero incremental cost, the relevant value is $F = 0$. When administrative coverage depends on the chosen program, the same construction yields $F(z)$, as studied in Proposition A.5. These costs are evaluated at common administrative prices across the technologies being compared. With a common fixed charge, a schedule containing joint approvals is weakly dominated by one joint package implementing its terminal allocation: balanced compensation supports that endpoint with one approval and no rented reserve. The two settlement rules thus exhaust the useful schedules within this authorization technology. Variable charges permit additional batching choices when those choices are included in the institutional menu.

### 4.1. *Executable productive releases and the boundary*

The geometric construction has a sufficient condition that does not depend on the coefficients of $M_h$. We state it for a finite linear economy with zero initial balances, positive values and costs, finite action caps, and reserve $\epsilon r$, where $r \gg 0$. Retain complete information, irreversible divisible adjustments, balanced compensation, and the $B \equiv 0$ mandate. Every demander–supplier pair and demander-only move is permitted. A voluntary bundle has at most one demander and one supplier coordinate, or a demander coordinate alone; a pair $(x, y)$ satisfies $v_i x \geq c_j y$. Assume a blocked start: no nonzero voluntary bundle is

feasible from zero. These assumptions define the quantitative class; the recovery theorem in the appendix also covers concave benefits and convex costs. Let $\mathcal{D}$ be the union of these voluntary directions normalized to have coordinate sum one, and put $\bar{v} = \max_i v_i$.

DEFINITION 3—An executable productive release: A feasible positive-surplus target $z^*$ admits an executable productive release if there are $\alpha, \kappa > 0$, nonzero voluntary bundles $u^1, \dots, u^k$, and a possibly empty finite list of nonzero voluntary bundles $v^1, \dots, v^\ell$, such that, writing

$$d = \sum_{j=1}^{k} u^j, \qquad p = Md, \qquad c = \sum_{j=1}^{\ell} v^j,$$

one has

$$p \geq 0, \quad p \neq 0, \quad z^* = \alpha d + c, \qquad M \sum_{j=1}^{s} u^j \geq -\kappa p \quad (0 \leq s \leq k),$$

and

$$\alpha p + M \sum_{j=1}^{s} v^j \geq 0 \quad (0 \leq s \leq \ell).$$

All inequalities between resource vectors are coordinatewise.

The vector $p$ is the net capacity released by one productive bundle. The prefix inequalities require that this same capacity can finance the bundle's temporary shortfalls. The final list then completes the target using the accumulated release. The prefix and completion inequalities distinguish an executable gain from a net release that cannot finance its own implementation. No positivity is required in every coordinate of $p$; resources with $p_i = 0$ must be generated before use within the specified bundle. Since all activities are nonnegative and the completed target respects the caps, every partial implementation also respects them.

THEOREM 4—A general logarithmic frontier and its boundary: *Suppose the preceding assumptions hold and $z^*$ admits an executable productive release. Put*

$$\sigma = \min_{i:p_i>0} \frac{r_i}{p_i} > 0, \qquad \Lambda = 1 + \frac{1}{\kappa} > 1.$$

*The minimum number $N^*(\epsilon)$ of voluntary approvals implementing $z^*$ with full reserve repayment satisfies*

$$N^*(\epsilon) \leq k \left\lceil \frac{\log(1+\alpha/(\sigma\epsilon))}{\log \Lambda} \right\rceil + \ell, \qquad N^*(\epsilon) = \Theta(\log(1/\epsilon)) \quad \textit{as } \epsilon \downarrow 0.$$

*In contrast, in a blocked-start economy possessing strictly positive weights $\lambda$ with*

$$\lambda^{\mathsf{T}} M d \leq 0 \quad \textit{for every normalized voluntary direction } d \in \mathcal{D},$$

*implementing any positive-surplus planner optimum with a fully repayable reserve requires $\Theta(1/\epsilon)$ approvals.*

PROOF SKETCH: From stock $sp$, the prefix inequalities permit a scaled productive bundle of size $s/\kappa$. Its release raises that stock to $\Lambda sp$. Starting with $\epsilon r \geq \sigma\epsilon p$, repeat this operation and truncate the last repetition when cumulative bundle size equals $\alpha$. Execute the completion list and return the reserve. The blocked-start bound in Theorem A.1supplies the matching logarithmic lower bound. Under nonexpansion, the weighted stock remains proportional to $\epsilon$, giving the inverse-reserve lower bound; divided execution gives the matching upper bound. Appendix C

provides the complete argument and verifies membership of every $M_h$, $h \in (0, 1/2)$. *Q.E.D.*

The nonexpansion and productive-execution conditions identify two distinct technological regimes. The former bounds the weighted stock available to each approval; the latter supplies an executable sequence that expands it and completes the planner allocation. Technologies outside these sufficient classes require a separate execution analysis. The coefficients of $M_h$ determine its exact frontier, while the general loss bounds depend on the resource and execution conditions.

### 4.2. *A sufficient test stated in technological primitives*

The executable-release condition can be certified without supplying a schedule in advance. The following finite collection of linear programs uses only $M$, values, costs, and action caps. It constructs a sufficient certificate for logarithmic implementation. Maintain the zero

baseline, blocked start, positive linear preferences, complete bilateral access, and $r \gg 0$ of Section 4.1. First solve the planner's linear program and denote its value by $V^* > 0$.

Let $\mathcal{T} = (D \times S) \cup \{(i,0) : i \in D\}$ be the finite set of bilateral and demander-only transaction types, and let $L = |D||S| + |D|$. For a pair $(i,j)$, define the cone

$$\mathcal{C}_{ij} = \{xe_i + ye_j : x, y \geq 0,\ v_i x \geq c_j y\}; \qquad \mathcal{C}_{i0} = \{xe_i : x \geq 0\}.$$

Fix a nonempty subset $R$ of resource accounts and a permutation $\pi$ of $\mathcal{T}$. Solve

$$\begin{aligned} L_{R,\pi} &:= \max \eta \\ \text{subject to } u^s &\in \mathcal{C}_{\pi(s)} \quad (s = 1, \ldots, L), \qquad z = \sum_{s=1}^{L} u^s, \\ 0 &\leq z \leq \bar{z}, \qquad W(z) = V^*, \qquad \eta \geq 0, \\ (Mz)_k &\geq \eta \quad (k \in R), \qquad (Mz)_k = 0 \quad (k \notin R), \\ \left(M \sum_{s=1}^{t} u^s\right)_k &\geq 0 \quad (k \notin R,\ t = 1, \ldots, L). \end{aligned}$$

Infeasible programs fail the test. The constraints permit a type to receive zero activity. They require the program to choose a planner optimum and divide it among voluntary transaction types. Accounts outside $R$ must be replenished before use; temporary shortfalls in $R$ can subsequently be funded by the positive release in those same accounts. This is a finite test on primitives, although the enumeration of supports and orders need not be computationally cheap.

PROPOSITION 5—A primitive certificate for logarithmic implementation: *If $L_{R,\pi} > 0$ for at least one support and ordering, an original planner optimum has a voluntary implementation with full reserve repayment and $\Theta(\log(1/\epsilon))$ approvals. More explicitly, take an optimal solution of that program, delete its zero bundles, and write $\nu$ for the number remaining and $p = Mz$. Define*

$$\kappa = \max_{k \in R,\, 1 \leq t \leq L} \frac{[-(M \sum_{s=1}^{t} u^s)_k]_+}{p_k}, \qquad \sigma = \min_{k \in R} \frac{r_k}{p_k}.$$

*Then $\kappa, \sigma > 0$, and the optimum is implementable with at most*

$$\nu \left\lceil \frac{\log(1 + 1/(\sigma\epsilon))}{\log(1 + 1/\kappa)} \right\rceil$$

*approvals. If a technology $M_C \leq M$ with the same preferences, caps, and trading permissions has a blocked start, positive planner value, and a strictly positive nonexpansion certificate, then Corollary 8 applies. Thus the primitive test also supplies a nonempty interval of common fixed commitment fees at sufficiently small approval prices at which the optimized institutional ranking reverses.*

PROOF SKETCH: The program itself supplies a planner optimum and its voluntary decomposition. Its positive value makes $p$ strictly positive exactly on $R$. The prefix inequalities outside $R$ and the definition of $\kappa$ give every prefix the lower bound $-\kappa p$. A blocked start rules out $\kappa = 0$. Hence the output satisfies Definition 3 with $d = z$, $\alpha = 1$, and no completion list. Apply Theorem 4 and then Corollary 8. Appendix Dsupplies details and verifies that the test passes for every $M_h$, $h \in (0, 1/2)$. *Q.E.D.*

A successful program constructs both the planner endpoint and its productive path. Allowing repeated transaction types or a separate completion stage enlarges the class of certificates beyond this finite test.

### 4.3. *An interacting three-stage release cycle*

The test also covers a technology with four accounts and three suppliers linked through one demander:

$$\widetilde{M}_h = \begin{pmatrix} -1 & 1 & 1 & 1 \\ 1 & -2 & -1 & h \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}, \qquad h \geq 0.$$

Let the demander value be one, each supplier cost be $1/10$, the demander cap be one, and each supplier cap be $1/3$. The first account implies $b_1 + b_2 + b_3 \geq a$, so the unique planner optimum and its value are

$$\widetilde{z}^* = (1, 1/3, 1/3, 1/3), \qquad \widetilde{V}^* = 9/10, \qquad \widetilde{M}_h \widetilde{z}^* = (0, h/3, 0, 0)^{\mathsf{T}}.$$

At this optimum every supplier cap binds. Each bilateral first move is blocked, including when $h > 0$. Three voluntary pairs with equal demander and supplier increments transfer usable capacity successively from account 2 to account 3, from 3 to 4, and from 4 to 2; the last transfer returns $1 + h$ units for each unit it consumes. The LP test passes with $R = \{2\}$ and order $(1, 2, 3)$ of suppliers. Its constants are $\kappa = 1/h$ and $\sigma = 3/h$ for the uniform reserve direction. This is an interacting resource cycle, with a shared demander and three suppliers, rather than a block-diagonal copy of the three-account example.

For a concrete optimized comparison, take $h = 0$ and $h = 1$, reserve price $\rho = 1$, approval cost $\tau = 10^{-6}$, and fixed joint fee $F = 1/5000$. The best joint value in either economy is $0.899799$. Under conservation the nonexpansion bound gives

$$\mathcal{V}_0^{\text{seq}} \leq \frac{9}{10} - \sqrt{\frac{3}{10^7}} < 0.899799.$$

At $h = 1$, reserve $10^{-6}$ and at most $57$ approvals implement the planner optimum with full repayment, giving

$$\mathcal{V}_1^{\text{seq}} \geq \frac{449971}{500000} = 0.899942 > 0.899799.$$

The allocations are freely chosen in both comparisons. The first inequality is a global upper bound and the second an attainable lower bound, which together establish the reversal without solving either sequential optimization problem exactly. All values, costs, and caps are unchanged across technologies. The complete primitive calculations are in Appendix D.

### 4.4. *Optimized welfare losses and institutional thresholds*

Institutional welfare depends on the allocation as well as its execution. We measure the loss after optimizing jointly over allocations, reserves, and schedules. For a zero-balance economy with planner value $V_M > 0$, define

$$\mathcal{V}_M^{\text{seq}}(\rho, \tau) = \max\left\{0, \sup_{(z,\epsilon,T) \text{ admissible}} [W(z) - \rho\epsilon - \tau T]\right\},$$

$$L_M(\rho, \tau) = V_M - \mathcal{V}_M^{\text{seq}}(\rho, \tau).$$

Admissibility includes employee-voluntary bilateral approvals, all action caps and intermediate account constraints, and full repayment of $\epsilon r$. The baseline incurs no implementation

cost. In particular, $0 \leq L_M \leq V_M$; the loss includes any reduction in gross surplus chosen to economize on implementation.

THEOREM 6—Optimized implementation losses: *Consider the positive-linear, complete-access class of Section 4.1, with zero original balances, $r \gg 0$, a blocked start, and $V_M > 0$. Fix $\rho > 0$. As $\tau \downarrow 0$:*

1. *If the technology admits a strictly positive nonexpansion certificate, then*

$$L_M(\rho, \tau) = \Theta(\sqrt{\rho\tau}).$$

2. *If a planner optimum admits an executable productive release in Definition 3, then*

$$L_M(\rho, \tau) = \Theta\big(\tau \log(\rho/\tau)\big).$$

*The constants depend on the fixed economy, its reserve direction, and the stated certificates. Both rates concern the optimized sequential value over all allocations, not merely the cost of a particular planner implementation. For all sufficiently small $\tau$, that optimized value is strictly positive.*

PROOF SKETCH: In the nonexpanding class, any implemented surplus $W > 0$ requires $T \geq aW/\epsilon$, where $a = \delta_\lambda/(\bar{v}S_\lambda) > 0$. Minimizing the resulting reserve and approval costs gives $2\sqrt{\rho\tau aW}$. The function $W - 2\sqrt{\rho\tau aW}$ is convex on $[0, V_M]$; its maximum occurs at an endpoint. Thus allowing the institution to reduce its allocation cannot remove the square-root lower bound. Divided execution of a planner optimum attains the matching order.

In the productive class, an allocation with $W < V_M/2$ already loses at least $V_M/2$ before implementation costs. For $W \geq V_M/2$, the blocked-start bound gives $T \geq \beta^{-1}\log(1 + c/\epsilon)$ for constants $\beta, c > 0$. Minimizing $\rho\epsilon + (\tau/\beta)\log(1 + c/\epsilon)$ gives a lower bound of logarithmic order. Executing the productive certificate at $\epsilon = \tau/\rho$ gives the matching upper bound. These feasible planner implementations have costs tending to zero, establishing positive adoption. Appendix Egives the bounds and the full proof. *Q.E.D.*

Optimized implementation loss determines the authority's willingness to pay for a common effectiveness instruction. Joint authorization saves the sequential loss but still requires

one ordinary approval. The remaining saving is the maximum additional program cost the authority will pay.

PROPOSITION 7—The joint-commitment threshold: *Let every nonzero joint package cost $\tau + F$, where $F \geq 0$, and keep the sequential costs above. If $\mathcal{V}_M^{\text{seq}}(\rho, \tau) > 0$, define*

$$F_M^*(\rho, \tau) = L_M(\rho, \tau) - \tau.$$

*Joint settlement has higher optimized net surplus when $F < F_M^*$; the institutions tie when $F = F_M^*$; sequential settlement has higher optimized net surplus when $F > F_M^*$. Under the blocked-start assumption, $F_M^* \geq \tau > 0$.*

*With $F = 0$, joint settlement weakly dominates sequential settlement for every $\rho, \tau > 0$. In the blocked-start class the dominance is strict whenever $V_M > \tau$. If instead $\mathcal{V}_M^{\text{seq}} = 0$, joint settlement is strictly preferred exactly when $F < V_M - \tau$; otherwise both institutions select the baseline.*

PROOF SKETCH: The joint value is $\max\{0, V_M - \tau - F\}$, whereas the sequential value is $V_M - L_M$. Comparing these expressions proves the threshold statements, including the baseline qualification. Every nonzero fully repayable schedule in a blocked-start economy requires at least two approvals. Hence $\mathcal{V}_M^{\text{seq}} \leq \max\{0, V_M - 2\tau\}$ and, when adoption is positive, $L_M \geq 2\tau$. More generally, a joint package with $F = 0$ implements any sequential endpoint with one approval and no reserve cost. Appendix Erecords the argument. *Q.E.D.*

COROLLARY 8—The interval supporting an endogenous reversal: *Compare a nonexpanding economy $C$ and an executable productive economy $P$ satisfying the respective hypotheses of Theorem 6, with common preferences, action caps, reserve direction, bilateral permissions, and $\rho > 0$. For all sufficiently small $\tau > 0$, both sequential values are positive and*

$$F_C^* = \Theta(\sqrt{\rho\tau}), \qquad F_P^* = \Theta\big(\tau \log(\rho/\tau)\big), \qquad \frac{F_P^*}{F_C^*} \longrightarrow 0.$$

*Every common fixed fee $F \in (F_P^*, F_C^*)$ makes joint settlement strictly preferable in $C$ and sequential settlement strictly preferable in $P$, after both institutions choose their allocations. If $M_P \geq M_C$ coordinatewise, this comparison is generated by a productive improvement*

*holding the other primitives fixed. For any fee $F > 0$ held fixed as $\tau \downarrow 0$, sequential settlement instead eventually dominates in both economies.*

PROOF SKETCH: Subtracting $\tau$ does not change either asymptotic order in Theorem 6. Their ratio tends to zero, so the fee interval is nonempty. Apply Proposition 7 to each economy. Both thresholds tend to zero, giving the last claim. *Q.E.D.*

The threshold makes the institutional choice operational. At a given approval price, the authority purchases joint authorization when its incremental resource cost is below the execution loss it avoids. Productive releases lower that willingness to pay. The common-fee interval therefore identifies programs for which an improvement in release capacity changes the chosen authorization technology. As ordinary approvals become arbitrarily cheap, both thresholds converge to zero and any fixed positive joint charge eventually favors sequential execution.

### 4.5. *Standing balances and the decision to borrow capacity*

Standing authorization capacity changes the cost of execution. To identify this margin, give the authority original balances $sr$, where $s \geq 0$ and $r \gg 0$ is the reserve direction. Original capacity may be consumed; only the additional reserve $\epsilon r$ must be returned. Zero rented reserve is admissible whenever the original ledger supports a schedule. Write $V(s)$ for the planner value and $L(s; \rho, \tau)$ for the optimized sequential loss relative to that value. The technology has a blocked start at $s = 0$. For $s > 0$, the authority can use its standing balances immediately.

The next result uses a condition on planner allocations over an interval $0 \leq s \leq \bar{s}$. A fixed voluntary core $d = \sum_{j=1}^{k} u^j$ has nonnegative release $p = Md$. For each $s$, a planner optimum can be written

$$z_s^* = \alpha_s d + \sum_{j=1}^{\ell_s} v_s^j, \qquad 0 < \underline{\alpha} \leq \alpha_s \leq \overline{\alpha} < \infty, \qquad \ell_s \leq \ell,$$

where the completion bundles are voluntary and satisfy

$$sr + \alpha_s p + M \sum_{j=1}^{t} v_s^j \geq 0 \qquad (0 \leq t \leq \ell_s).$$

We call this a uniform core and completion certificate. It describes technologies in which standing capacity changes the amount assigned to a common executable core and finances a bounded number of terminal adjustments. It is a finite condition on activity bundles and account balances. When the bundles and $\alpha_s$ are affine in $s$, feasibility can be checked at the two endpoints of the interval; a common planner dual solution verifies optimality throughout it. Appendix F

states this finite verification precisely.

THEOREM 9—Implementation losses with standing balances: *Retain the positive-linear, complete-access assumptions of Section 4.1, a blocked start at zero balances, $V(0) > 0$, and a uniform core and completion certificate. Fix $\rho > 0$. Uniformly over $s \in [0, \bar{s}]$, as $\tau \downarrow 0$:*

1. *If the technology has a strictly positive nonexpansion certificate, then*

$$L_C(s;\rho,\tau) \asymp \Phi_C(s;\rho,\tau) := \inf_{\epsilon \geq 0} \left\{ \rho\epsilon + \frac{\tau}{s+\epsilon} \right\} \asymp \frac{\tau}{s + \sqrt{\tau/\rho}}.$$

2. *If the core has $p \neq 0$ and satisfies $M \sum_{j=1}^{t} w^j \geq -\kappa p$ for every prefix and some $\kappa > 0$, then*

$$\begin{aligned} L_P(s;\rho,\tau) &\asymp \Phi_P(s;\rho,\tau) \\ &:= \inf_{\epsilon \geq 0} \left\{ \rho\epsilon + \tau \log\left(1 + \frac{1}{s+\epsilon}\right) \right\} \\ &\asymp \tau \log\left(1 + \frac{1}{s+\tau/\rho}\right). \end{aligned}$$

*Here $\asymp$ denotes two-sided bounds with positive constants independent of $s$ and sufficiently small $\tau$. The constants may depend on the fixed economy, the certificates, $\bar{s}$, and $\rho$.*

PROOF SKETCH: The initial usable stock is $(s+\epsilon)r$. The nonexpansion and blocked-start arguments bound approvals from below using this total stock. The bounds apply to every allocation with surplus at least $V(0)/2$; choosing a lower-surplus allocation incurs a fixed welfare loss. For the upper bounds, execute the core using total stock $s+\epsilon$, by divided execution under conservation and geometric expansion under productivity. The completion inequalities preserve the entire rented reserve. These constructions attain the planner optimum, so their losses contain no allocation approximation error. Minimization over $\epsilon$ gives the displayed envelopes. Appendix Fsupplies the uniform bounds. *Q.E.D.*

Put $\eta = \tau/\rho$. The conserving envelope is exactly

$$\Phi_C = \begin{cases} 2\sqrt{\rho\tau} - \rho s, & s \leq \sqrt{\eta}, \\ \tau/s, & s \geq \sqrt{\eta}. \end{cases}$$

The productive envelope is minimized at total capacity $\max\{s, t_P\}$, where $t_P = (\sqrt{1+4\eta} - 1)/2 \sim \eta$. These are minimizers of the bounding envelopes. The following implementation decisions follow from the matching bounds themselves.

TABLE II

STANDING CAPACITY AND OPTIMIZED IMPLEMENTATION LOSS

| Standing capacity | Conserving technology | Productive technology |
|---|---|---|
| $s \lesssim \eta$ | Loss $\asymp \sqrt{\rho\tau}$; borrowing of order $\sqrt{\eta}$ attains this rate. | Loss $\asymp \tau \log(1/\eta)$; borrowing of order $\eta$ attains this rate. |
| $\eta \ll s \ll \sqrt{\eta}$ | Loss $\asymp \sqrt{\rho\tau}$; any implementation attaining this order must borrow capacity of order $\sqrt{\eta}$. | Loss $\asymp \tau \log(1/s)$; existing balances suffice to attain this order. |
| $\sqrt{\eta} \lesssim s \to 0$ | Loss $\asymp \tau/s$; existing balances suffice. | Loss $\asymp \tau \log(1/s)$; existing balances suffice. |
| Fixed $s > 0$ | Loss $\asymp \tau$, with a coefficient of order $1/s$ as $s$ becomes small. | Loss $\asymp \tau$, with a coefficient of order $\log(1/s)$ as $s$ becomes small. |

The middle regime identifies a financing decision: productive releases allow the authority to implement its planner allocation using standing capacity, whereas conservation requires additional authorization to attain the optimal welfare-loss order. At fixed positive balances, both technologies' sequential losses become linear in the approval price. Their coefficients retain the economically relevant distinction.

COROLLARY 10—Institutional choice with positive standing balances: *Consider a conserving economy and a productive economy satisfying the respective conditions of Theorem 9, with common preferences, action caps, trading permissions, $s, \rho, \tau$, and reserve direction. For every path $s = s(\tau) \to 0$, including strictly positive paths,*

$$L_P(s; \rho, \tau)/L_C(s; \rho, \tau) \longrightarrow 0.$$

*Moreover, for every sufficiently small fixed $s > 0$, there is a nonempty interval of constants $f_s > 0$ such that the common joint commitment fee $F = f_s\tau$ makes joint settlement optimal in the conserving economy and sequential settlement optimal in the productive economy, for all sufficiently small $\tau$. Each institution chooses its allocation. The same conclusion applies to a technological improvement when preferences, caps, and trading permissions are common and the two technologies additionally satisfy $M_C \leq M_P$.*

PROOF SKETCH: The envelopes imply the ratio limit. For fixed $s$, uniform constants give $L_C \geq c_C\tau/s$ and $L_P \leq C_P\tau\log(1+1/s)$ for small $\tau$. As $s \downarrow 0$, the first coefficient exceeds the second. Choose

$$\max\{0, C_P\log(1+1/s) - 1\} < f_s < c_C/s - 1$$

and apply the threshold $F^* = L - \tau$. Planner values remain bounded away from zero, ensuring adoption. The full proof is in Appendix F. *Q.E.D.*

## 5. EXACT COSTS, ENDOGENOUS ALLOCATIONS, AND ROBUSTNESS

### 5.1. *The exact cost of a proportional target*

For $0 \leq h \leq 1/4$, put

$$Q_h(\theta;\rho,\tau) = \inf_{\epsilon>0}\{\rho\epsilon + \tau N_h(\theta,\epsilon)\}.$$

This is a cost conditional on implementing $z_\theta$. The endogenous welfare comparison below allows each institution to choose any feasible allocation.

THEOREM 11—The exact cost frontier: *For $0 \leq h \leq 1/4$, $\theta \in (0,1]$, and $\rho, \tau > 0$,*

$$Q_h(\theta;\rho,\tau) = \min_{T\in\{2,3,\ldots\}}\{\rho\epsilon_T(h,\theta) + \tau T\}. \tag{5}$$

*An optimal reserve and integer approval count exist. The reserve infimum can equivalently be taken over $0 < \epsilon \leq \theta/4$. For each fixed $h > 0$ and $\theta > 0$, as $\tau/\rho \downarrow 0$, $Q_h = \Theta(\tau\log(\rho/\tau))$, whereas $Q_0 = \Theta(\sqrt{\rho\theta\tau})$.*

PROOF SKETCH: At each approval count the exact frontier gives the least reserve; minimizing over integers proves the formula. Its objective tends to infinity with $T$, so a minimum

exists. Two equal pair increments implement the target from reserve $\theta/4$, making larger reserves unnecessary. At $h = 0$, minimize $\rho\theta/[12(T-1)] + \tau T$. At fixed $h > 0$, the exact reserve decreases geometrically in $T$; a count proportional to $\log(\rho/\tau)$ attains the upper order, and the geometric lower bound gives the matching order. Appendix Bsupplies the full argument. *Q.E.D.*

### 5.2. *Reversal when each institution chooses its allocation*

Let $\mathcal{V}_h^J$ and $\mathcal{V}_h^{\text{seq}}$ denote the best joint and sequential employee surplus net of implementation costs, including the zero-cost baseline. The joint value is

$$\mathcal{V}_h^J = \max\{0, V_h^* - F - \tau\}.$$

The sequential value is the supremum of $W(z) - \rho\epsilon - \tau T$ over all voluntary admissible schedules, all $\epsilon > 0$, and all terminal allocations whose reserve can be fully returned, also including zero. Sequential settlement can select every feasible endpoint and its admissible compensated schedule.

THEOREM 12—Endogenous-allocation institutional reversal: *In the family $M_h$, fix $\rho = 12$, $\tau = 1/10000$, $\mathcal{K} = 0$, and charge $F = 1/100$ for every nonzero joint package. Then*

$$\mathcal{V}_0^J = \frac{8899}{10000} > \frac{8799}{10000} = \mathcal{V}_0^{\text{seq}},$$

*whereas*

$$\mathcal{V}_{1/4}^{\text{seq}} \geq \frac{27280647}{30064375} > \frac{98889}{110000} = \mathcal{V}_{1/4}^J.$$

*Thus, under the same fixed implementation costs, productive releases reverse the welfare ranking of joint and sequential settlement even when each institution may choose its allocation. The productive lower bound is attained by a voluntary 16-approval schedule implementing $z_{1/4}^*$ with reserve $1/144309$ in every account and full repayment.*

PROOF SKETCH: At $h = 0$, repayment and conservation imply that every feasible terminal allocation is $z_\theta$, $0 \leq \theta \leq 1$. For $T \geq 2$, the exact frontier makes net surplus at most

$$\frac{9}{10}\theta - \frac{\theta}{T-1} - \frac{T}{10000}.$$

For each $T$, this is linear in $\theta$, so a positive optimum uses $\theta = 1$. Writing $n = T - 1$, the minimum of $1/n + (n+1)/10000$ is $201/10000$, attained at $n = 100$. The best joint value is $9/10 - 101/10000$.

At $h = 1/4$, Proposition 1 gives the global planner value $10/11$. Proposition 2 implements $z_{8/11}$ with 16 approvals and reserve $e = 1/144309$. The target releases $1/11$ of pension capacity. A final supplier-2 approval with demander increment $3/11$ and supplier increment $2/11$ consumes that release, adds $1/11$ of position capacity, and reaches $(1, 4/11, 6/11)$. It has positive surplus and leaves the reserve repayable. Its implementation cost is

$$\overline{C} = \frac{12}{144309} + \frac{16}{10000} = \frac{50603}{30064375} < \frac{101}{10000}.$$

The last first-stage approval and the completion use the same pair. Combining them preserves linear feasibility and balanced voluntary compensation, so only 16 approvals are charged. Hence its net value exceeds the best joint value, whose allocation is also chosen optimally. Appendix Gverifies every approval, transfer, cap, and account balance. The productive sequential number is an attainable lower bound, which suffices for the optimized ranking. *Q.E.D.*

TABLE III

ENDOGENOUS ALLOCATIONS AND THE REVERSAL OF INSTITUTIONAL WELFARE

| Technology | Best joint | Best sequential | Higher value |
|---|---|---|---|
| Conserving ($h = 0$) | 0.889900 | 0.879900 | Joint |
| Productive ($h = 1/4$) | 0.898991 | $\geq \underline{V}_P$ | Sequential |

The productive lower bound is $\underline{V}_P = 27280647/30064375 \simeq 0.907408$. Decimals are rounded; the inequality refers to the exact fraction. Both joint entries and the conserving sequential entry are global optimized values. The productive sequential entry uses a feasible 16-approval schedule. Preferences, caps, prices, and the fixed fee on every nonzero joint package are unchanged.

The exact productive frontier and cost theorem concern proportional targets. They provide the first stage of this proof, while the endogenous planner calculation supplies the correct endpoint. Theorem 12 therefore strengthens the fixed-target comparison in the online appendix by implementing the endogenous planner optimum. The appendix displays the 17-stage construction and the merged final approval explicitly. The feasible witness establishes

the productive lower bound required for the ranking; both joint values and the conserving sequential value are exact.

Corollary A.3in the online appendix gives the corresponding fixed-target comparison and its exact cost-minimizing reserves.

### 5.3. *When standing capacity changes the institution*

The three-account family supplies an explicit certificate. For $0 \le h < 1/2$ and $0 \le s \le 1/32$, original balances $s\mathbf{1}$ give

$$z^*_{h,s} = \left(1, \frac{1-8s}{2+3h}, \frac{1+2h+(4+2h)s}{2+3h}\right), \qquad V^*_{h,s} = 1 - \frac{1+h}{5(2+3h)} + \frac{2-h}{5(2+3h)}s.$$

Implement the proportional target with $\theta_s = 2(1-8s)/(2+3h)$ using total initial capacity $s+\epsilon$. One supplier-2 approval then adds

$$(\Delta a, \Delta b_2) = \left(\frac{3h+16s}{2+3h}, \frac{2h+(12+2h)s}{2+3h}\right).$$

It reaches $z^*_{h,s}$, consumes the permitted original balances, and returns $\epsilon$ in each account. Compensation equals $\Delta b_2/10$, leaving the supplier indifferent and the demander strictly better off whenever the completion is nonzero.

PROPOSITION 13—An exact frontier with standing capacity: *In $M_0$ with original balances $s\mathbf{1}$, $0 \le s \le 1/32$, the minimum additional reserve implementing the planner optimum in at most an integer $T \ge 2$ voluntary approvals is*

$$\epsilon^*_T(s) = \left[\frac{1-8s}{12(T-1)} - s\right]_+.$$

*In particular, for $s > 0$ the minimum number of approvals implementing that optimum without rented capacity is*

$$N_0(s) = \left\lceil 1 + \frac{1-8s}{12s} \right\rceil.$$

PROOF SKETCH: Let $t = s + \epsilon$ and let $q$ be the difference between payroll and position balances. Conservation bounds $|q| \le 3t$ at every prefix. Initially $q = 0$; at the planner endpoint $q = -3s$. Each supplier's activity equals twice its contribution to the variation of

$q$. Hence total supplier activity satisfies $1 - 2s \leq 12t(T-1) + 6s$, giving the lower bound. Alternating boundary transactions attain this variation, with a final supplier-2 transaction that leaves the full reserve available for repayment. When standing balances exceed the required amplitude, smaller alternating excursions attain the same endpoint without renting capacity. Every transaction has positive employee surplus. Appendix F.5gives the construction and proves optimality over all schedules. *Q.E.D.*

The formula separates two uses of original capacity. A larger stock directly replaces rented capacity, and it changes the planner's supplier mix, reducing the quantity that must pass through repeated conversions. Because approvals are discrete, institutional choice can include an interval of indifference rather than a single cutoff.

COROLLARY 14—Standing capacity determines when productivity changes the institution: *Fix* $\rho = 12$*,* $\tau = 1/10000$*,* $F = 1/100$*, and original balances* $s\mathbf{1}$*, with* $0 \leq s \leq 1/32$*. Define*

$$s_J = \frac{1}{1208}, \qquad s_S = \frac{9901}{11960000}.$$

*After optimizing allocations, reserves, and schedules, the conserving economy* $M_0$ *strictly prefers joint settlement for* $s < s_J$*, is indifferent for* $s \in [s_J, s_S]$*, and strictly prefers sequential settlement for* $s > s_S$*. The productive economy* $M_{1/4}$ *strictly prefers sequential settlement throughout the interval* $[0, 1/32]$.

PROOF SKETCH: For $n = T - 1 \geq 2$, the conserving sequential optimum conditional on $n$ uses the planner allocation and has implementation cost

$$C_n(s) = \left[\frac{1 - 8s}{n} - 12s\right]_+ + \frac{n+1}{10000}.$$

Schedules with one or two approvals, including their optimal quantity choices, cannot improve on joint settlement below the stated cutoff. The $n = 100$ schedule first matches the joint cost $101/10000$ at $s_J$; the $n = 99$ schedule first matches it at $s_S$ and falls below it for larger stocks. Every other approval count crosses later or has an approval cost already exceeding the joint cost. In $M_{1/4}$, the proportional construction and one completion reach the new planner optimum using at most 17 approvals and rented reserve at most $1/144309$.

Their cost is strictly below $101/10000$ for every stated $s$. Appendix F.5verifies the global optimization and the cutoff inequalities. *Q.E.D.*

This comparison holds prices and commitment costs fixed while varying standing authorization capacity. Productivity changes the chosen institution when $s < s_J$. Once $s > s_S$, existing capacity already makes sequential settlement preferable under conservation, so productivity improves its performance without changing the institution. The interval between these cutoffs reflects the integer number of approvals.

For the same prices as Theorem 12, take $s = 1/2500$. In the productive economy, nine proportional approvals and the completion implement the planner optimum using only original balances. Table IV reports exact joint values, the exact conserving optimum, and this attainable productive value. The fixed fee is $F = 0.01$ in both economies. Joint productive value is rounded to six decimals; its exact value is $494473/550000$. The productive sequential lower bound is rounded downward from $249739/275000$. The two conserving entries are exact.

TABLE IV

OPTIMIZED INSTITUTIONAL COMPARISON WITH POSITIVE ORIGINAL BALANCES

| Technology | Best joint value | Best sequential value | Preferred rule |
|---|---|---|---|
| $h = 0$ | 0.889980 | 0.884812 | Joint |
| $h = 1/4$ | 0.899042 | $\geq 0.908141$ | Sequential |

All accounts initially contain positive capacity. Productivity changes the authority's institution and removes the need to rent capacity in the displayed planner implementation. Appendix F

verifies the fractions, compensation, intermediate balances, and global optimization; Appendix F.5proves the exact conserving value.

*The conserving optimum.* Proposition A.4gives the full conserving solution. With common verification cost $\mathcal{K}(z_\theta) = k\theta$, the optimal positive quantity is one. The authority compares the joint fee with $G(\rho, \tau) = \min_{n \geq 1}\{\rho/(12n) + \tau n\}$. The minimizing integer satisfies $\tau n(n-1) \leq \rho/12 \leq \tau n(n+1)$; the sequential schedule uses $n+1$ approvals and reserve

$1/(12n)$. This exact choice connects the resource frontier to the inventory balance in Baumol (1952).

*Allocation and participation margins.* The comparison also permits the joint fee to vary with the allocation. Proposition A.5proves that every common fee profile in $[0.005, 0.015]$ preserves the reversal; a coverage charge can change the joint quantity to $0.99z_h^*$. Proposition A.6

shows that concave benefits can reduce the sequentially chosen quantity to $7/8$, and that employee participation can raise the two-approval reserve above its physical minimum. The appendix gives the complete institutional optimizations and compensated schedules.

*Fixed adjustment units and original slack.* The reversal also holds with a fixed adjustment unit and positive original account balances. The following extension imposes the same activity grid on both institutions; the core model remains divisible. Let $\mathcal{V}_{h,\delta}^{J}(R^0)$ and $\mathcal{V}_{h,\delta}^{\mathrm{seq}}(R^0)$ be their optimized net values when every activity increment is a multiple of $\delta$, original balances are $R^0$, and every rented reserve is returned. Transfers and reserve capacity remain divisible, and the authority selects the allocation and schedule.

COROLLARY 15—Reversal with fixed units and original account slack: *Maintain the prices and financing assumptions of Theorem 12. Fix* $\delta = 1/500$ *and let* $R^0 \in [0, 1/2500]^3$. *Then*

$$\mathcal{V}_{0,\delta}^{J}(R^0) = \frac{8899}{10000} > \frac{44399}{50000} \geq \mathcal{V}_{0,\delta}^{\mathrm{seq}}(R^0),$$

*whereas*

$$\mathcal{V}_{1/4,\delta}^{\mathrm{seq}}(R^0) \geq \frac{9013}{10000} > \frac{8989}{10000} = \mathcal{V}_{1/4,\delta}^{J}(R^0).$$

*The productive lower bound uses a 17-approval schedule with reserve* $1/2000$ *per account, implements a grid-constrained planner optimum, and returns that reserve in full. The result includes every strictly positive original ledger in the displayed box.*

*Proof sketch.* An exact grid schedule implements $(1, 183/500, 272/500)$, with gross surplus $909/1000$. Its reserve and approval costs total $77/10000$. Feasibility is preserved when original balances increase. In the conserving economy, a variation bound on the difference between payroll and position balances gives the global upper bound $\mathcal{V}_0^{\mathrm{seq}}(s\mathbf{1}) \leq$

$8799/10000 + (101/5)s$, even with divisible activities. Planner dual bounds and the activity grid give the two exact joint values. Monotonicity in the original ledger extends the bounds to the entire box. Appendix Jsupplies the schedule and proof. The specified grid and slack box therefore preserve the strict institutional ranking at the common implementation prices.

## 6. AN EMPIRICAL APPLICATION: PRIORITY IN WATER ALLOCATION TRADE

Administrative records from the Victorian Water Register show how processing order changes the allocation of a positive stock of authorized trade opportunity. The application measures successful requests and executed water volume under alternative priority rules. These outcomes make the authority's allocation decision observable in an institution with binding capacity. Requests are treated as indivisible at their submitted volumes, connecting the exercise to Appendix L.

### 6.1. *Institution and administrative replay*

For the October 2025 Goulburn to Murray opening, the authority randomized applications submitted during a common window. The Australian Competition and Consumer Commission independently observed the randomization (ACCC, 2025). The published application table contains 203 eligible requests, submitted on 15 October and processed on 16 October. Each row records the requested volume, processing order, outcome, and trade opportunity before and after processing. Receipt identifiers also recover arrival order (DEECA, 2025a, Table 5 and notes).

The stock available immediately before this queue was $R = 31{,}215.1$ megalitres (ML). It equals the opening opportunity of $31{,}523.4$ ML, less $326.3$ ML of earlier tagged use, plus $18.0$ ML of reverse trade (DEECA, 2025a). The reverse trade replenished usable corridor capacity before the queue began. It therefore enters the starting stock held fixed across rules. No tagged use occurred during the queue. The common pool contains the 203 eligible applications; it excludes two ineligible submissions and a later trade that used the final $5.3$ ML.

Let $v_i > 0$ be the submitted volume and $q$ a permutation of the pool $\mathcal{I}$. The capacity benchmark accepts a request in full whenever it fits, and continues after a refusal:

$$Y_i(q) = \mathbf{1}\left\{ v_i \le R - \sum_{j \prec_q i} v_j Y_j(q) \right\}, \qquad A(q) = \sum_i Y_i(q), \qquad Q(q) = \sum_i v_i Y_i(q).$$

Here $A$ counts successful applications and $Q$ measures executed volume. Exact arithmetic reproduces every observed October decision and every intermediate balance. Appendix Mdocuments the 391 records covering October and December, the exclusions, and the replay.

### 6.2. *Priority, application counts, and capacity use*

Write $v_{(1)} \le \cdots \le v_{(203)}$ for the ordered request sizes. The maximum application count under the capacity benchmark is

$$A^*(R) = \max\left\{ k : \sum_{j=1}^{k} v_{(j)} \le R \right\}.$$

Processing the smallest requests first attains this bound: every subset of $k$ requests consumes at least the sum of the $k$ smallest volumes. The maximum need not select a unique order. In these data,

$$\sum_{j=1}^{159} v_{(j)} = 31{,}193.0 \;\le\; 31{,}215.1 \;<\; 31{,}693.0 = \sum_{j=1}^{160} v_{(j)}.$$

Thus $159$ is the exact maximum, and executing $160$ requests would require at least another $477.9$ ML of trade opportunity under this benchmark.

The change from recorded priority to smallest first yields 61 more successful applications, a $62.2\%$ increase relative to 98. The gap is $38.4\%$ of the attainable count. With the stated tie rule, 79 observed successes remain selected, 19 are displaced, and 80 previously refused requests enter. Executed volume falls by only $16.8$ ML. Near exhaustion of the volume limit therefore coexists with a substantial difference in how many applications receive capacity. A simulation of 100,000 uniform permutations gives a mean of $93.66$ successes and a central $95\%$ range of 75 to 112. The recorded 98 successes are typical of this conditional lottery benchmark.

TABLE V

OCTOBER WATER TRADE: THE SAME APPLICATIONS AND STARTING CAPACITY

| Processing rule | Successful applications | Volume (ML) | Unused (ML) |
|---|---|---|---|
| Recorded random priority | 98 | 31,209.8 | 5.3 |
| Arrival order | 106 | 31,206.4 | 8.7 |
| Smallest volume first | 159 | 31,193.0 | 22.1 |

The first row reproduces the administrative record. The other rows are capacity counterfactuals for the same 203 submitted requests, assuming all other requirements remain satisfied. All rules process 203 applications. Smallest first breaks equal volumes by recorded priority. Source: DEECA (2025a, Table 5); calculations in Appendix M.

### 6.3. *Economic interpretation and account requirements*

The application identifies an allocation margin: priority determines which requests use existing capacity. A count objective places equal weight on applications. A welfare objective instead requires their gains from trade; 191 of the 203 requests are classified as related party trades, and 174 record a zero price. These entries do not identify those gains. Applications also need not represent distinct traders. A permanent change in priority could change submission sizes and application splitting, whereas the reported comparison holds the submitted menu fixed.

The comparison clarifies the role of settlement in the theory. In the scalar capacity benchmark, any feasible selected subset can be executed sequentially in any order. Joint selection and an optimal sequential rule consequently attain the same count. The measured gap concerns the particular priority rule. The main theorems additionally characterize the cost of execution when separate accounts and productive releases constrain the authority's chosen allocation. Moreover, the 203 administrative decisions are distinct from the successful application counts in Table V; those counts cannot be substituted for the costly approval count in the theory.

December provides a direct observation of an additional account requirement. At priority 45, an application for $62.3$ ML was refused with $23{,}824.0$ ML of trade opportunity remaining. The report attributes the refusal to an insufficient seller balance (DEECA, 2025b, Table 5, note 8). The seller account is a private entitlement alongside the public trade limit. Its unobserved links and balance history determine whether a different schedule could fund that seller. The October count maximum is therefore conditional on the other account

requirements. Together, the records establish a substantial allocation margin and distinct execution constraints.

The $18.0$ ML reverse trade illustrates how completed transactions replenish capacity for later use. This addition enters the initial stock $R$ held fixed in our comparisons. The productive-release theorem requires a voluntary sequence that finances its intermediate steps, supports repeated expansion, and permits completion at the chosen planner allocation. The records establish an operational channel for replenishment; the theorem identifies when that channel changes optimized implementation costs and institutional choice.

## 7. CONCLUSION

Productive releases change the cost of implementing a planner-optimal allocation under separate public accounts. Conserving transactions repeatedly use existing authorization capacity, whereas an executable productive sequence creates capacity for subsequent approvals. The exact frontier and the general loss bounds quantify this distinction after the authority chooses both its allocation and its schedule.

Standing balances determine how the mechanism enters institutional decisions. With empty accounts, conservation and productivity generate different small-cost rates. With fixed positive balances, both losses are linear in the approval price, but the required approval counts depend inversely and logarithmically, respectively, on scarce standing capacity. The uniform comparison identifies a range in which productive implementation attains the minimum-loss order using existing balances, while conserving implementation gains from temporary reserves. Productive releases therefore reduce the standing capacity needed to economize on paid authorization and joint commitment. The commitment thresholds translate this quantitative advantage into a reversal of the preferred settlement rule, with allocations endogenous under both institutions. The exact conserving frontier identifies a standing-capacity cutoff above which both technologies select sequential settlement at the same fixed implementation prices.

The water trade application makes the allocation margin concrete. With positive initial trading opportunity and fixed submitted requests, different priority rules produce nearly identical volume use and sharply different application counts. Tracking the selected allocation and the underlying account constraints is therefore essential to evaluating an execution rule.

The theory supplies the further comparison when productive releases and separate public accounts determine the cost of implementing that allocation.

The framework preserves the distinction between employee compensation and public authorization. Transfers secure participation; resource releases finance execution. The main results retain complete information and the $B \equiv 0$ mandate. The appendix extends the analysis to valued public balances and discrete decisions, identifying the financing conditions and reserve floors relevant to those environments. These results provide a basis for comparing account reforms through their effects on standing capacity, temporary authorization, and approval requirements.

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